
Gaussian mollification of exact LCHS kernels: optimal precision scaling inside a minimal single-QSP selector

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Abstract

The linear combination of Hamiltonian simulations (LCHS) writes the non-unitary propagator of a dissipative linear differential equation as a weighted Fourier integral of Hamiltonian evolutions. Its cost is set almost entirely by the bandwidth $k_{\max}$ of that integral, which in turn is fixed by the choice of kernel. Kernels that satisfy the LCHS contour identity exactly cannot do better than $k_{\max} = \mathcal{O}(\log^{1/\beta} \epsilon^{-1})$ with $\beta < 1$; kernels that satisfy it only approximately can reach the optimal $\mathcal{O}(\log \epsilon^{-1})$. We show that the optimal class is generated from the near-optimal one by a single, elementary operation: multiplication by the entire mollifier $M(k) = \exp[c(1 - ik) - (1 + k^2)/4\gamma^2]$, which acts in the time variable as a shifted Gaussian smoothing of the exponential decay. The mollifier preserves the LCHS normalisation for every parameter value, converts any admissible exact kernel into an approximate one with a closed-form erfc error bound, and reproduces two published optimal kernels as the special cases in which the exact seed is the constant or the Cauchy kernel. Applying it to the An–Childs–Lin kernel gives a new (β, γ, c) family. We show that the resource that actually binds in an explicit circuit is not the bandwidth but the ℓ^1 norm of the weights, which fixes the success probability, and that at a matched ℓ^1 budget the achievable bandwidth obeys $k_{\max} \simeq 2 \log(\epsilon^{-1}) / \log \alpha_{\max}$. All kernels are implemented inside an unmodified minimal LCHS circuit — sinusoidal (Fejér–Clenshaw–Curtis) quadrature, a single quantum-signal-processing selector and the sparse block-encodings of the Hermitian parts — and run end to end on a state-vector emulator of a fault-tolerant quantum computer for the periodic advection–diffusion equation. Across 509 emulated circuits the mollified kernels reach a given accuracy with a median of $2.0\times$ and up to $4\times$ fewer block-encoding queries than the An–Childs–Lin baseline at equal or higher success probability, the measured precision law changes from $\epsilon \sim e^{-bk_{\max}^{0.66 \pm 0.03}}$ to $\epsilon \sim e^{-bk_{\max}^{1.14 \pm 0.04}}$, and the Fejér–Clenshaw–Curtis accuracy floor at a fixed Fourier register drops by up to $4.9\times$ — except at the very smallest register, where the substitution is a net loss and we say why.

1 Introduction

Many equations of practical interest — advection–diffusion in fluid and plasma transport, the Fokker–Planck and Liouville equations with dissipation, linear embeddings of nonlinear dynamics such as the Koopman–von Neumann and Carleman constructions — take the form

$$\partial_t \psi = -A\psi, \quad \psi(t) = e^{-At} \psi(0), \quad (1)$$

with a non-Hermitian generator A . A quantum computer applies unitaries, so the non-unitary propagator e^{-At} must be dilated. The *linear combination of Hamiltonian simulations* (LCHS) of An,

Liu and Lin [1] does this by an integral over an auxiliary Fourier coordinate,

$$e^{-At} = \int_{\mathbb{R}} \frac{\xi(k)}{1 - ik} e^{-i(A_H + kA_L)t} dk, \quad A = A_L + iA_H, \quad (2)$$

where $A_L = (A + A^\dagger)/2 \succeq 0$ and $A_H = (A - A^\dagger)/(2i)$ are Hermitian. The identity holds for any *kernel* ξ that decays, is continuous, is analytic in the lower half plane and is normalised by $2\pi \xi(-i) = 1$; the last three conditions make the contour in (2) closable around the pole at $k = -i$.

On a quantum computer (2) is truncated to $|k| \leq k_{\max}$, discretised on N_k nodes, and compiled as a linear combination of unitaries whose select operator performs the N_k Hamiltonian simulations. The dominant cost, the number of queries Q_A to the block-encoding of A , is proportional to $\|A_H + k_{\max}A_L\|t$ and therefore *linear in* $k_{\max}$. Everything about the precision–cost trade-off of LCHS is thus the question: how fast can the weight $w(k) = \xi(k)/(1 - ik)$ be made to decay?

An, Childs and Lin [2] answered this for kernels that satisfy (2) *exactly*: a no-go argument forbids $|w| \sim e^{-\mathcal{O}(|k|)}$, and their family

$$\xi_\beta(k) = \frac{1}{2\pi} e^{2\beta} \exp[-(1 + ik)^\beta], \quad \beta \in (0, 1), \quad (3)$$

attains the best possible exact behaviour, $\epsilon \sim e^{-\mathcal{O}(k_{\max}^\beta)}$, i.e. $k_{\max} = \mathcal{O}(\log^{1/\beta} \epsilon^{-1})$. The obstruction is removed if the identity (2) is only required to hold to accuracy ϵ : such *approximate-LCHS* kernels, constructed by analytic extension of the decaying exponential [7, 5, 6], achieve the optimal $k_{\max} = \mathcal{O}(\log \epsilon^{-1})$.

Independently of the kernel, the *circuit* matters. Novikau and Joseph [3] showed that substituting $k = k_{\max} \sin \theta$ turns the discretised integral into Fejér–Clenshaw–Curtis (FCC) quadrature and, more importantly, lets a *single* quantum-signal-processing (QSP) circuit perform all N_k Hamiltonian simulations in parallel with $\mathcal{O}(1)$ ancillae, which is markedly cheaper than the multiplexed block-encodings used elsewhere. Their implementation, however, uses the near-optimal kernel (3); whether an optimal-class kernel actually pays inside that circuit, at the resolutions an emulator can reach, was left open.

Contributions.

1. **A generating operation for optimal kernels.** We show (Sec. 2) that multiplying *any* admissible exact kernel ξ_0 by the entire mollifier $M(k; \gamma, c) = \exp[c(1 - ik) - (1 + k^2)/4\gamma^2]$ yields an approximate-LCHS kernel that (i) automatically satisfies the LCHS normalisation, because $M(-i) = 1$; (ii) has the closed-form error bound $\sup_{x \geq 0} |F(x) - e^{-x}| \leq \frac{1}{2} \operatorname{erfc}(\gamma c - \frac{1}{2\gamma}) S_0$; and (iii) inherits Gaussian weight decay $e^{-k^2/4\gamma^2}$. The mollifier acts in the time variable as a shifted Gaussian smoothing of the step function that the exact kernel reproduces. Two kernels from the recent optimal-LCHS literature [5] are exactly the cases $\xi_0 = 1/2\pi$ and $\xi_0 = \xi_{\text{sp}}$ of this construction; taking $\xi_0 = \xi_\beta$ gives a new (β, γ, c) family.
2. **The binding resource is the ℓ^1 norm, not the bandwidth.** The LCU sub-normalisation $\alpha = \|w\|_1$ fixes the success probability. We show that the shift c buys bandwidth at exactly the rate $k_{\max} \simeq 2 \log(\epsilon^{-1})/c$ while costing $\alpha \simeq \alpha_0 e^c$, so at a matched α budget the optimal kernel is the one whose *exact seed* has the smallest ℓ^1 norm. This explains a non-obvious ordering of the families that we then observe on the emulator.
3. **An end-to-end implementation and measurement.** We add a kernel-agnostic weight-construction subroutine to a minimal FCC/single-QSP LCHS circuit, compute and verify QSVT angles for the kernel-dependent weight functions, and run 509 circuits on a state-vector emulator. The mollified kernels need a median of $2.0\times$ and up to $4\times$ fewer A -oracle queries than the An–Childs–Lin baseline at matched end-to-end accuracy, at equal or higher success probability, and the fitted precision law changes from $k_{\max}^{0.66}$ to $k_{\max}^{1.14}$, i.e. the query exponent in $\log \epsilon_{\text{LCHS}}^{-1}$ falls from 1.52 ± 0.06 to 0.88 ± 0.03 .
4. **A resolution map.** We measure which family is cheapest at which $(n_k, \epsilon_{\text{LCHS}})$, and report a second, previously unremarked benefit: because the mollified weight vanishes to all orders at the truncation boundary in the same Gaussian factor, the Fejér–Clenshaw–Curtis accuracy

floor at a fixed Fourier register drops by up to $4.9\times$. And we report where the picture is *not* favourable — that the naive log–log exponent over the emulator-accessible precision window is misleading, and that at the smallest register we tried ($n_k = 5$) the substitution is a net loss.

2 Kernels

2.1 Exact kernels and the obstruction

Because A_L and A_H are Hermitian, (2) reduces to a scalar statement. Write

$$F(x) := \int_{\mathbb{R}} w(k) e^{-ikx} dk, \quad w(k) = \frac{\xi(k)}{1 - ik}, \quad (4)$$

so that (2) holds iff $F(x) = e^{-x}$ for all $x = \lambda t$ in the spectrum of $A_L t$, i.e. for all $x \geq 0$. For a kernel analytic in $\Im k < 0$ the contour may be closed below and only the pole of $(1 - ik)^{-1}$ at $k = -i$ contributes, giving $F(x) = 2\pi\xi(-i)e^{-x}$; the normalisation condition is exactly $2\pi\xi(-i) = 1$. Conversely, if $|w|$ decayed exponentially then F would be analytic in a strip around the real axis, hence equal to e^{-x} for all x by continuation, contradicting $|F| \leq \|w\|_1$. This is the obstruction that caps exact kernels at $|w| \sim e^{-b|k|^\beta}$, $\beta < 1$, realised by (3) with $b = \cos(\beta\pi/2)$.

2.2 Gaussian mollification

Proposition 1 (Mollification). *Let ξ_0 be an admissible exact LCHS kernel with weight $w_0 = \xi_0/(1 - ik)$, $F_0 = \widehat{w_0}$, $S_0 := \sup_{x \leq 0} |F_0(x)| < \infty$, and let*

$$M(k; \gamma, c) = \exp\left[c(1 - ik) - \frac{1 + k^2}{4\gamma^2}\right], \quad \gamma > 0, c > 0. \quad (5)$$

Then $\xi = \xi_0 M$ satisfies $2\pi\xi(-i) = 1$, is entire wherever ξ_0 is, $|\xi(k)| = |\xi_0(k)| e^c e^{-(1+k^2)/4\gamma^2}$, and its weight function obeys

$$\sup_{x \geq 0} |F(x) - e^{-x}| \leq \frac{1}{2} \operatorname{erfc}\left(\gamma c - \frac{1}{2\gamma}\right) (1 + S_0 e^{2c}). \quad (6)$$

Proof sketch. $M(-i) = \exp[0 - 0] = 1$ because both $1 - ik$ and $1 + k^2 = (1 - ik)(1 + ik)$ vanish at $k = -i$; the normalisation and the modulus are immediate. In the time variable, multiplication by M is convolution with $\mathcal{G}(y) = \frac{\gamma}{\sqrt{\pi}} e^{c-1/4\gamma^2} e^{-\gamma^2(y+c)^2}$, a Gaussian of width $1/(\sqrt{2}\gamma)$ centred at $y = -c$ whose exponential moment is exactly one, $\int e^y \mathcal{G}(y) dy = 1$. Hence for $x \geq 0$

$$F(x) = \int F_0(x - y) \mathcal{G}(y) dy = e^{-x} \int_{y < x} e^y \mathcal{G}(y) dy + \int_{y > x} F_0(x - y) \mathcal{G}(y) dy,$$

using $F_0(u) = e^{-u}$ for $u \geq 0$. The first term is e^{-x} minus the Gaussian tail beyond $y = x \geq 0$, which is bounded by $\frac{1}{2} \operatorname{erfc}(\gamma c - \frac{1}{2\gamma})$; the second is bounded by $S_0 e^{2c}$ times the same tail. $\square$

For the two simplest seeds the convolution can be done in closed form. With $\xi_0 = 1/2\pi$ (the “flat” seed, which satisfies the residue identity but does not decay on its own) and with the Cauchy kernel $\xi_{\text{sp}} = 1/[\pi(1 + ik)]$ one obtains

$$\xi^{(1)}(k) = \frac{1}{2\pi} e^{c(1-ik)} e^{-(1+k^2)/4\gamma^2}, \quad F^{(1)}(x) = \frac{1}{2} e^{-x} \operatorname{erfc}\left(\frac{1}{2\gamma} - \gamma(x + c)\right), \quad (7)$$

$$\xi^{(2)}(k) = \frac{e^{c(1-ik)}}{\pi(1 + ik)} e^{-(1+k^2)/4\gamma^2}, \quad F^{(2)}(x) = e^c [f_1(x + c) + f_1(-x - c)], \quad (8)$$

with $f_1(u) = \frac{1}{2} e^{-u} \operatorname{erfc}(\frac{1}{2\gamma} - \gamma u)$. These are precisely the kernels $\hat{f}_1$ and $\hat{f}_2$ of Low and Somma [5], whose $\hat{f}_{j,y}$ family corresponds to the rational seeds $\xi_0 = \frac{1}{2\pi} \left(\frac{y+1}{y+ik}\right)^{j-1}$; (7)–(8) are their equations (6) and (53) in the present normalisation. We verified (7)–(8) against direct numerical Fourier integration to ten decimal places, and found the bound (6) to be attained at $x = 0$ to fourteen digits (proposal/code/kernel_derivation/verification.json).

Numerical verification of the construction. We checked all of the above numerically (proposal/code/kernel_derivation/verify_kernel.py). The normalisation $|2\pi\xi(-i) - 1|$ is 0 to machine precision for every seed and every (γ, c) ; the rational and exponential seeds are finite everywhere in $\Im k < 0$ on a dense 401×101 grid; the closed forms (7)–(8) agree with direct numerical Fourier integration to ten decimal places; the bound (6) is attained at $x = 0$ to fourteen digits (e.g. $\gamma = 10, c = 0.27$: measured supremum 8.9244×10^{-5} against the bound 8.9244×10^{-5}); and the truncated tail $\int_{|k|>R} |w|$ follows $e^{-R^2/4\gamma^2}$ with a slowly varying algebraic prefactor over ten orders of magnitude.

The new member. Nothing in Proposition 1 requires the seed to be rational. Taking the An–Childs–Lin kernel (3) as the seed gives the three-parameter family used as the headline variant of this work,

$$\xi_{\beta,\gamma,c}(k) = \frac{1}{2\pi} \exp\left[2^\beta - (1+ik)^\beta + c(1-ik) - \frac{1+k^2}{4\gamma^2}\right] \quad (9)$$

whose weight decays as $\exp[-\cos(\beta\pi/2)|k|^\beta - k^2/4\gamma^2]$: the exact factor supplies extra decay at moderate k at no cost in the approximation error, and the mollifier supplies the asymptotically optimal decay. The family degenerates continuously to the An–Childs–Lin kernel as $\gamma \rightarrow \infty, c \rightarrow 0$ and to (7) as $\beta \rightarrow 0$.

2.3 What actually limits the bandwidth

Truncating at $k_{\max}$ costs $\epsilon_{\text{trunc}} = \int_{|k|>k_{\max}} |w| dk$, which for the mollified family is $\mathcal{O}(\gamma^2 k_{\max}^{-3} e^c) e^{-k_{\max}^2/4\gamma^2}$. Together with (6) the two error sources balance at $\gamma c \simeq \sqrt{L}$ and $k_{\max} \simeq 2\gamma\sqrt{L}$ with $L = \log \epsilon^{-1}$, i.e.

$$k_{\max} \simeq \frac{2L}{c}. \quad (10)$$

So the shift c — and nothing else — sets the constant in the optimal $\mathcal{O}(\log \epsilon^{-1})$ law. But c is not free. The block-encoded LCHS sum is a linear combination of unitaries with sub-normalisation $\alpha = \sum_j |w_j|$, and the success probability of the circuit is $\|\psi(t)\|^2/\alpha^2$; from $|\xi| = |\xi_0| e^c e^{-(1+k^2)/4\gamma^2}$ we get $\alpha \rightarrow \alpha_0 e^c$ as $\gamma \rightarrow \infty$, where $\alpha_0 = \|w_0\|_1$ is the seed’s own ℓ^1 norm. Hence, at a fixed one-norm budget $\alpha \leq \alpha_{\max}$,

$$k_{\max} \gtrsim \frac{2\log(\epsilon^{-1})}{\log(\alpha_{\max}/\alpha_0)}. \quad (11)$$

Equation (11) is the practical content of the optimal-kernel construction: the prefactor of the optimal scaling is inversely proportional to the logarithm of the *spare* one-norm, and the best seed is the one that leaves the most spare. The Cauchy kernel is extremal here — its weight $1/[\pi(1+k^2)]$ is positive, so $\alpha_0 = 1$ exactly and the entire budget is available — whereas the An–Childs–Lin seed already spends $\alpha_0 \approx 1.31$ at $\beta = 0.7$. This is why, in Sec. 5, the β -seeded family (9) wins at loose and intermediate precision (where its extra $e^{-b|k|^\beta}$ decay dominates) while the Cauchy-seeded kernel (8) takes over at the tightest precisions (where only the shift matters).

2.4 Discretisation and weights

We keep the sinusoidal discretisation of [3] unchanged: $\theta_j = -\pi/2 + j\Delta\theta$, $\Delta\theta = \pi/(N_k - 1)$, $k_j = k_{\max} \sin \theta_j$, $j = 0, \dots, N_k - 1$ with $N_k = 2^{n_k}$, and

$$w_j = \frac{1}{\mathcal{N}} k_{\max} \cos(\theta_j) \Delta\theta \frac{\xi(k_j)}{1-ik_j}, \quad \mathcal{N} = \sum_j k_{\max} \cos(\theta_j) \Delta\theta \frac{\xi(k_j)}{1-ik_j}. \quad (12)$$

The division by $\mathcal{N}$ enforces the LCHS normalisation $\int \xi(k)/(1-ik) dk = 1$ *exactly on the finite grid*. It is applied identically to every kernel including the baseline, for which it is a no-op to within quadrature error; for the mollified kernels it removes a systematic amplitude defect $F(0) \neq 1$ that would otherwise dominate the comparison.

Table 1: QSVT angles for the weight function $\sqrt{w}$, $n_k = 8$, tolerance 10^{-4} . “QSP vs. polynomial” is an independent re-evaluation of the phase sequence; “QSP vs. target” additionally includes the Chebyshev truncation.

kernel	parity	deg.	#ang.	QSP vs. poly.	QSP vs. target
ACL $\beta=0.7$, $k_{\max}=53.99$	even	422	423	1.1×10^{-12}	8.8×10^{-5}
	odd	491	492	1.5×10^{-12}	6.1×10^{-6}
mollified $\beta=0.7$, $k_{\max}=29.27$	even	238	239	1.9×10^{-13}	4.1×10^{-5}
	odd	587	588	1.3×10^{-12}	2.1×10^{-5}
mollified Cauchy $\hat{f}_2$, $k_{\max}=24.33$	even	418	419	4.7×10^{-13}	9.1×10^{-5}
	odd	759	760	8.8×10^{-13}	1.9×10^{-6}

Because A_L and A_H commute for the uniform advection–diffusion generator used below, the discretised LCHS operator is diagonal in the same basis as A , and its operator-norm error is exactly

$$\eta = \max_{x \in [0, T]} \left| \sum_j w_j e^{-ik_j x} - e^{-x} \right|, \quad T = \|A_L\|t. \quad (13)$$

η bounds the relative state-vector error for *any* initial condition, so we use it — and never a property of the test state — to select $(\gamma, c, k_{\max})$. Every number reported as an accuracy is nevertheless the emulator-measured state-vector error.

3 Circuit

We hold the implementation framework of [3] fixed and change only the kernel and its weight-construction subroutine. Specifically we retain: the sinusoidal/FCC quadrature; the selector realised as one QSP circuit of N_O multiplexed qubitization iterates acting on the block-encoding U_C of $C_j = A_H + \sin(\theta_j)k_{\max}A_L$; the two QSP ancillae; and the sparse block-encodings U_{A_H}, U_{B_m} of the Hermitian parts, including their state-preparation, $\sin \theta$ and incrementer/decrementer sub-circuits. No cost is externalised to an oracle: every reported gate and query count is produced by the compiled circuit.

Kernel-agnostic weight construction. The reference build hard-codes the An–Childs–Lin weights in a direct-decomposition (multiplexed $R_y R_z$) profile. We added one profile that implements the whole family $\xi = (2\pi)^{-1} \left(\frac{y+1}{y+ik} \right)^m e^{2^\beta - (1+ik)^\beta} M(k; \gamma, c)$ with parameters $(k_{\max}, m, y, \beta, \gamma, c)$ and the normalisation of (12), switching the mollifier off for $\gamma \leq 0$ so that the same code path reproduces the baseline bit-for-bit. Running the weight oracle alone on the emulator and comparing the N_k output amplitudes with $\sqrt{w_j}/\sqrt{N_k}$ gives a maximum absolute deviation of 2.2×10^{-15} (baseline), 4.8×10^{-15} (variant) and 4.0×10^{-16} (Low–Somma $\hat{f}_2$).

QSVT angles for the weight function. The weight depends on the Fourier coordinate only through $z = \sin \theta = k/k_{\max}$, which the $\sin \theta$ sub-circuit block-encodes exactly as $\text{diag}(\sin \theta_j)$; a QSVT circuit on that block-encoding therefore realises any polynomial $P(z)$. Since $w(-k) = \overline{w(k)}$ for every kernel considered here, $\sqrt{w} = A(z) + iB(z)$ with A even and B odd on a continuous branch of the square root, so the weight oracle is a linear combination of one even and one odd QSVT circuit. For each kernel we expand A and B in Chebyshev polynomials, truncate at a stated tolerance, obtain the angles with an independent L-BFGS symmetric-QSP solver, and then *re-evaluate the QSP product independently* to check it. Table 1 reports the result. This construction is not used in the headline runs, which follow the reference build in using the direct decomposition (it needs one ancilla instead of three); it is reported because the weight-construction cost is kernel-dependent and must be accounted for.

4 Experimental setup

Test problem. The one-dimensional periodic advection–diffusion equation $\partial_t \psi = -v \partial_x \psi + D \partial_x^2 \psi$ with $v = 1.0$, $D = 0.01$ on $x \in [0, 1)$, discretised on $N_x = 2^6$ points with $\Delta x = 1/(N_x - 1)$,

and a Gaussian initial condition of width 0.05 centred at $x = 0.5$ — the test problem of [3]. Then $\|A_L\| = 158.76$ and $\|A_H\| = 63.0$. We simulate to $t = 0.0125$, i.e. $\|A_L t\| = 1.98$ and $\|A_H t\| = 0.79$.

Disclosure. The reference paper uses $t = 0.8$, for which $\|A_L t\| = 127$ and the FCC quadrature needs $n_k \geq 10$. We are interested here in the regime where the Fourier register is small — $n_k \in \{5, \dots, 8\}$, which is where the choice of kernel is most consequential and where a state-vector emulation of the full circuit is comfortable — and the shorter time is what puts that regime in reach. Nothing else about the test problem is changed.

Selector compilation and the $k_{\max}$ grid. The QSP-Hamiltonian phase sequence is taken from the angle set shipped with the reference implementation ($\tau_b = 39.25$, $\epsilon_{\text{QSP}} = 10^{-12}$, $M_b = 75$ qubitization queries per repetition) and repeated n_t times, so that $Q_A = n_t M_b$ and $\tau = n_t \tau_b$. The block-encoding sub-normalisation is $\Lambda = \frac{\zeta}{2}(\|A_H\| + k_{\max}\|A_L\|)$ with $\zeta = 4$, the tightest value compatible with the branch amplitudes of the reference prepare circuit, and $\tau = 2\Lambda t$; requiring τ to be an exact multiple of τ_b quantises $k_{\max}$ on the grid $k_{\max}(n_t) = (n_t \tau_b / 4t - \|A_H\|) / \|A_L\| = \{4.55, 9.49, 14.44, \dots\}$. Both the compilation overhead $M_b / \tau_b = 1.91$ and the quantisation apply identically to every kernel, so ratios and fitted exponents are unaffected; absolute query counts carry the 1.91 factor and we say so wherever they are quoted.

Metrics. ϵ_{LCHS} is the relative state-vector error $\|\psi^{\text{circ}}(t) - \psi^{\text{exact}}(t)\| / \|\psi^{\text{exact}}(t)\|$, with the exact state computed at run time by dense matrix exponentiation and the circuit output rescaled only by the analytically known LCU and amplitude-amplification gains. Q_A counts applications of U_C ; each contains one U_{A_H} and one U_{B_m} . STMC is the single-target multi-controlled gate count of the entire circuit reported by the emulator. The success probability is the squared norm of the zero-ancilla output block. Ancillae are all qubits except r_x , including the k -register.

Pre-registration and parameter selection. The kernel form, the axis, the test problem, the metrics and the sweep grid were fixed in `proposal/code/kernel_plan.{md,json,py}` before any validation emulation was run. (γ, c) and $k_{\max}$ are then selected by minimising $k_{\max}$ subject to $\eta \leq \eta_{\text{target}}$ in (13) and to the one-norm budget $\alpha \leq \alpha_{\max} = 1.3053$, which is the An-Childs-Lin $\beta = 0.7$ baseline’s own supremum over the declared grid (the exact kernels are never capped). All families in the declared list are reported, including those on which the variant loses.

Baseline fidelity. Re-running the reference implementation’s own input for $(n_x, n_k, k_{\max}, t) = (6, 7, 20, 0.8)$ reproduces its logged gate count $N_{\text{gates}} = 2\,692\,135$ exactly and gives success probability 0.2151 (the paper quotes ~ 0.2). With the analytically known sub-normalisations, the emulator output agrees with our classical model of the discretised LCHS operator to 1.1×10^{-10} relative, after accounting for the reference build’s rounding of the QSP time argument from 2489.49 to 2489. All baseline numbers quoted below are from our own runs of that implementation, not from the paper.

5 Results

5.1 Precision scaling

Figure 1(a) shows, for every query budget on the declared grid at $(n_x, n_k) = (6, 8)$, the best end-to-end accuracy each family achieves. The mollified kernels lie below the exact ones at every budget. Comparing at matched accuracy over all 24 baseline points that a mollified kernel can match, the query reduction is between $1.0\times$ and $3.5\times$ (median $2.0\times$), always at equal or higher success probability.

Figure 1(b) and Table 3 give the precision law. Fitting $\log \epsilon_{\text{LCHS}}^{-1} = a + b k_{\max}^q$ to the emulator-measured error of a *single fixed kernel* swept over the whole $k_{\max}$ grid gives $q = 0.85 \pm 0.11$ for the An-Childs-Lin $\beta = 0.7$ baseline ($n = 12$ points spanning $\epsilon_{\text{LCHS}} \in [2.1 \times 10^{-5}, 1.8 \times 10^{-2}]$), $q = 1.35 \pm 0.35$ for the mollified Cauchy kernel and $q = 1.05 \pm 0.50$ for the mollified $\beta = 0.7$ kernel. Because Q_A is affine in $k_{\max}$, the query exponent is $p = 1/q$, so these read $p = 1.18 \pm 0.15$ against $p = 0.74 \pm 0.19$ and $p = 0.95 \pm 0.45$.

The emulator fits are limited by how many grid points each kernel spans before reaching its own floor, and by the fact that some curves are not monotone: the An-Childs-Lin $\beta = 0.8$ kernel, for instance, returns $q = 1.28 \pm 0.22$, larger than its own theoretical β , because its error is flat over

Table 2: Primary results, all measured on the QuCF state-vector emulator for the periodic advection–diffusion equation with $n_x = 6$, $v = 1$, $D = 0.01$, $t = 0.0125$. ϵ_{LCHS} is the relative state-vector error against run-time classical propagation; Q_A counts block-encoding queries (each containing one U_{A_H} and one U_{B_m}); STMC is the single-target-multi-controlled gate count of the whole circuit (weights, amplitude amplification, selector, initialisation); “anc.” is every qubit except r_x , including the k -register. Absolute Q_A carries the common 1.91 QSP-compilation factor of Sec. 4. Rows are the For $n_k = 6$, the headline configuration, the complete measured envelope of each family is listed; for $n_k = 8$ its cheapest, median and most accurate points are. “ADE” is the advection–diffusion test problem. The exact kernels are the $\gamma \rightarrow \infty$, $c = 0$ limit of the mollified family, for which the mollifier is the identity.

Method	Test	(n_x, n_k, t)	$k_{\max}$	(γ, c)	Q_A	STMC	ϵ_{LCHS}	anc.	p_{succ}
ACL $\beta=0.7$ (baseline)	ADE	(6, 6, 0.0125)	4.55	$(\infty, 0)$	75	16821	1.80×10^{-2}	18	0.846
ACL $\beta=0.7$ (baseline)	ADE	(6, 6, 0.0125)	9.49	$(\infty, 0)$	150	31676	1.09×10^{-2}	18	0.644
ACL $\beta=0.7$ (baseline)	ADE	(6, 6, 0.0125)	14.44	$(\infty, 0)$	225	46531	3.30×10^{-3}	18	0.570
ACL $\beta=0.7$ (baseline)	ADE	(6, 6, 0.0125)	19.38	$(\infty, 0)$	300	61094	2.80×10^{-3}	18	0.477
ACL $\beta=0.7$ (baseline)	ADE	(6, 6, 0.0125)	24.33	$(\infty, 0)$	375	75949	1.33×10^{-3}	18	0.476
ACL $\beta=0.7$ (baseline)	ADE	(6, 6, 0.0125)	29.27	$(\infty, 0)$	450	90804	5.95×10^{-4}	18	0.473
ACL $\beta=0.7$ (baseline)	ADE	(6, 6, 0.0125)	39.16	$(\infty, 0)$	600	120514	5.13×10^{-4}	18	0.467
mollified $\beta=0.7$ (ours)	ADE	(6, 6, 0.0125)	4.55	(2.00, 0.375)	75	16821	3.31×10^{-3}	18	0.601
mollified $\beta=0.7$ (ours)	ADE	(6, 6, 0.0125)	9.49	(4.25, 0.075)	150	31676	7.89×10^{-4}	18	0.654
mollified $\beta=0.7$ (ours)	ADE	(6, 6, 0.0125)	14.44	(4.25, 0.150)	225	46239	7.75×10^{-4}	18	0.476
mollified $\beta=0.7$ (ours)	ADE	(6, 6, 0.0125)	19.38	(5.50, 0.100)	300	61094	2.67×10^{-4}	18	0.478
mollified $\beta=0.7$ (ours)	ADE	(6, 6, 0.0125)	34.22	(11.50, 0.025)	525	105659	2.39×10^{-4}	18	0.475
mollified Cauchy $\hat{f}_2$	ADE	(6, 6, 0.0125)	4.55	(2.12, 0.575)	75	16529	3.33×10^{-3}	18	0.477
mollified Cauchy $\hat{f}_2$	ADE	(6, 6, 0.0125)	9.49	(4.00, 0.375)	150	31676	4.88×10^{-4}	18	0.609
mollified Cauchy $\hat{f}_2$	ADE	(6, 6, 0.0125)	14.44	(5.88, 0.266)	225	46531	3.39×10^{-4}	18	0.675
ACL $\beta=0.7$ (baseline)	ADE	(6, 8, 0.0125)	4.55	$(\infty, 0)$	75	28097	1.80×10^{-2}	20	0.874
ACL $\beta=0.7$ (baseline)	ADE	(6, 8, 0.0125)	34.22	$(\infty, 0)$	525	119759	5.05×10^{-4}	20	0.553
ACL $\beta=0.7$ (baseline)	ADE	(6, 8, 0.0125)	58.94	$(\infty, 0)$	900	197034	2.10×10^{-5}	20	0.552
ACL $\beta=0.8$	ADE	(6, 8, 0.0125)	4.55	$(\infty, 0)$	75	28097	3.68×10^{-3}	20	0.736
ACL $\beta=0.8$	ADE	(6, 8, 0.0125)	39.16	$(\infty, 0)$	600	134146	1.14×10^{-4}	20	0.392
ACL $\beta=0.8$	ADE	(6, 8, 0.0125)	63.88	$(\infty, 0)$	975	211421	8.55×10^{-6}	20	0.392
mollified $\beta=0.7$ (ours)	ADE	(6, 8, 0.0125)	4.55	(2.00, 0.375)	75	27029	3.30×10^{-3}	20	0.590
mollified $\beta=0.7$ (ours)	ADE	(6, 8, 0.0125)	29.27	(8.25, 0.050)	450	104304	9.16×10^{-6}	20	0.560
mollified $\beta=0.7$ (ours)	ADE	(6, 8, 0.0125)	49.05	(11.25, 0.025)	750	166124	1.89×10^{-6}	20	0.562
mollified Cauchy $\hat{f}_2$	ADE	(6, 8, 0.0125)	4.55	(2.12, 0.575)	75	27029	3.33×10^{-3}	20	0.555
mollified Cauchy $\hat{f}_2$	ADE	(6, 8, 0.0125)	24.33	(6.25, 0.350)	375	88849	2.70×10^{-5}	20	0.562
mollified Cauchy $\hat{f}_2$	ADE	(6, 8, 0.0125)	39.16	(9.12, 0.266)	600	135214	6.74×10^{-6}	20	0.616

the first four budgets and then drops sharply, which a single power law reads as a large exponent. The same law measured on the kernels themselves — the operator-norm error on a dense $k_{\max}$ grid with the quadrature made non-limiting — is far better resolved: $q = 0.659 \pm 0.026$ for the An–Childs–Lin $\beta = 0.7$ kernel (theory: $q = \beta = 0.7$), $q = 0.732 \pm 0.021$ for $\beta = 0.8$ (theory 0.8), and $q = 1.140 \pm 0.042$ and $q = 1.161 \pm 0.085$ for the two mollified kernels (theory: $q = 1$ for the envelope at fixed shift, $q = 2$ for the pure Gaussian factor alone). The separation between the exact and the mollified class is 10σ . In query terms the baseline’s exponent is $p = 1.52 \pm 0.06$ and the mollified kernels’ $p = 0.88 \pm 0.03$, i.e. the substitution moves the query complexity from the near-optimal $\mathcal{O}(\log^{1/\beta} \epsilon_{\text{LCHS}}^{-1})$ towards the optimal $\mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$, as predicted.

Headline configuration. At $(n_x, n_k) = (6, 6)$ — 24 qubits in total, 18 of them ancillary — the An–Childs–Lin $\beta = 0.7$ baseline reaches $\epsilon_{\text{LCHS}} = 2.80 \times 10^{-3}$ with $Q_A = 300$ block-encoding queries and success probability 0.477. The mollified $\beta = 0.7$ kernel of this work reaches 7.89×10^{-4} — 3.5 times more accurate — with $Q_A = 150$ and success probability 0.654: $2\times$ fewer queries at strictly better accuracy and a 37% higher success probability. At the baseline’s *best* achievable accuracy at that resolution, 5.13×10^{-4} ($Q_A = 600$), the mollified Cauchy kernel needs $Q_A = 150$ — 4 times fewer — and the mollified $\beta = 0.7$ kernel goes on to 2.39×10^{-4} , 2.1 times better than

Table 3: Precision law $\log \epsilon_{\text{LCHS}}^{-1} = a + b k_{\text{max}}^q$; the query complexity exponent is $p = 1/q$ in $Q_A \propto \log^p \epsilon_{\text{LCHS}}^{-1}$. “emulator” fits the measured end-to-end state-vector error over the whole declared k_{max} grid at $(n_x, n_k) = (6, 8)$, reduced to the running minimum over the grid; “kernel” fits the operator-norm error of the same kernel on a dense k_{max} grid with the quadrature made non-limiting ($n_k = 12$), and is the precise determination of the kernel’s own truncation law.

kernel	q (emulator)	n	q (kernel)	$p = 1/q$ (kernel)
ACL $\beta=0.7$ (baseline)	0.85 ± 0.11	12	0.659 ± 0.026	1.52 ± 0.06
ACL $\beta=0.8$	1.28 ± 0.22	9	0.732 ± 0.021	1.37 ± 0.04
Cauchy	0.52 ± 0.03	13	0.079 ± 0.001	12.67 ± 0.19
mollified Cauchy, $\gamma=11.5, c=0.20$	1.35 ± 0.35	8	1.140 ± 0.042	0.88 ± 0.03
mollified $\beta=0.7, \gamma=6.5, c=0.075$	1.05 ± 0.50	5	1.161 ± 0.085	0.86 ± 0.06

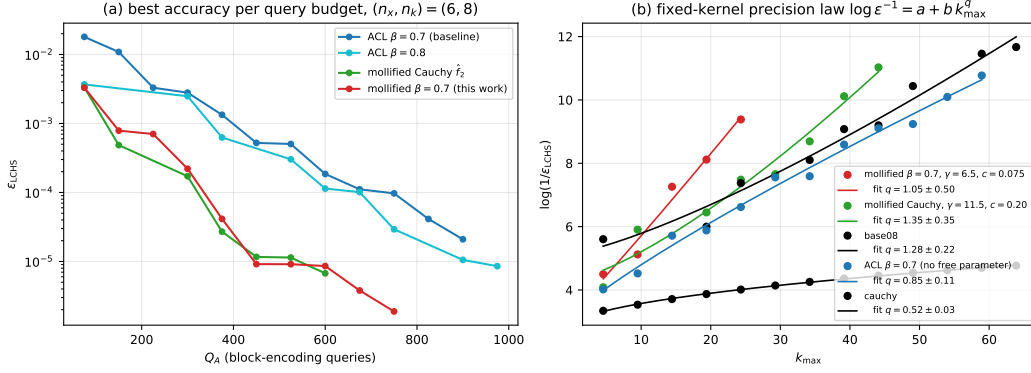


Figure 1: Precision scaling at $(n_x, n_k) = (6, 8)$, $t = 0.0125$. (a) best emulator-measured accuracy each family reaches at each query budget on the declared grid; every point is a separate emulated circuit. (b) the fixed-kernel precision law: the five kernels of Table 3, each with *no* parameter re-tuned between points, swept over the whole k_{max} grid and reduced to the running minimum, together with the fitted $\log \epsilon_{\text{LCHS}}^{-1} = a + b k_{\text{max}}^q$. The exact kernels’ stretched exponentials ($q < 1$ for the Cauchy and $\beta = 0.7$ kernels) are visibly flatter than the mollified kernels’ ($q > 1$).

anything the baseline can do there. Across all 253 mollified runs the success probability never falls below 0.47, well above the ~ 0.2 of the reference implementation’s published configuration.

5.2 Kernel-family characterisation

Table 4 reports, for each Fourier-register size and target accuracy, the smallest query budget at which each family reaches it. The mollified kernels are never more expensive, and are cheaper by $1.0\text{--}2.5\times$ in the cells where the target is tight enough that the quantisation of k_{max} does not already put every family at the smallest admissible bandwidth. Within the mollified class the ordering follows Sec. 2.3: the β -seeded kernel of this work and the Cauchy-seeded $\hat{f}_2$ are within one grid step of each other over most of the map, with $\hat{f}_2$ ahead at the tightest accuracies, where the whole one-norm budget is best spent on the shift, and the β -seeded kernel ahead where its extra $e^{-b|k|^{0.7}}$ factor still pays for itself.

Table 5 shows why the tighter cells are empty at small n_k : every family eventually reaches a Fejér–Clenshaw–Curtis floor beyond which more bandwidth makes the answer *worse*, because the N_k nodes can no longer resolve the weight function. Those floors are themselves strongly kernel-dependent, and the mollified kernels reach a deeper one at $n_k = 6$ ($1.7\times$), $n_k = 7$ ($4.9\times$), $n_k = 8$ ($4.5\times$). The exception is $n_k = 5$, where only 32 nodes are available: there the mollified kernels are quadrature-starved before their own truncation error matters and the exact kernels floor $1.7\times$ lower (Sec. 5.4). At $n_k = 6$ the An–Childs–Lin $\beta = 0.7$ kernel bottoms out at 5.13×10^{-4} while the mollified $\beta = 0.7$ kernel reaches 2.39×10^{-4} , a factor of 2.1 deeper; the mechanism is analysed in Sec. 5.3.

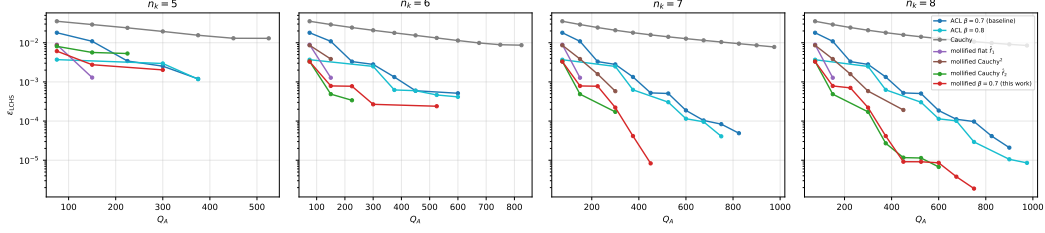


Figure 2: Kernel-family characterisation. Emulator-measured error against A -oracle queries for every family in the declared list, at each Fourier-register size. At $n_k = 5$ and 6 the exact kernels hit a quadrature floor that the mollified kernels do not; at $n_k = 8$ all families resolve their own truncation error and the ordering is set by the weight decay.

Table 4: Kernel-family characterisation: smallest A -oracle query budget at which each family reaches the stated emulator-measured accuracy. “gain” is the ratio of the best exact kernel to the best mollified kernel. “ACL” is the An–Childs–Lin kernel at the stated β ; “m-” denotes its Gaussian-mollified counterpart. An entry > 975 means the accuracy was not reached by that family at any $k_{\max}$ on the declared grid, whose largest budget is 975 queries. Combinations that no kernel reaches at that resolution are omitted; see Table 5 for the corresponding quadrature floors.

n_k	ϵ_{LCHS}	ACL 0.7	ACL 0.8	Cauchy	m-Cauchy $\hat{f}_2$	m-0.7 (ours)	gain
5	10^{-2}	225	75	> 975	75	75	1.00
6	10^{-2}	225	75	675	75	75	1.00
6	10^{-3}	450	375	> 975	150	150	2.50
7	10^{-2}	225	75	825	75	75	1.00
7	10^{-3}	450	375	> 975	150	150	2.50
7	10^{-4}	750	675	> 975	> 975	375	1.80
8	10^{-2}	225	75	825	75	75	1.00
8	10^{-3}	450	375	> 975	150	150	2.50
8	10^{-4}	750	750	> 975	375	375	2.00

5.3 Mechanism

Three mechanisms combine to produce the measured gain.

(i) *Weight decay.* The mollifier multiplies the weight by $e^c e^{-k^2/4\gamma^2}$. At a fixed γ the truncation error therefore falls *super*-exponentially in $k_{\max}$ — faster even than the optimal asymptotic law — and it is only the need to increase γ as the target precision tightens (to keep the mollification error (6) down) that degrades the envelope to the exponential $k_{\max} \simeq 2L/c$ of (10). At emulator-accessible precisions one is largely in the first, super-exponential regime, which is why the measured gains (median $2.0\times$, up to $4\times$) exceed what the asymptotic laws alone would suggest at these ϵ .

(ii) *The one-norm budget selects the seed.* Equation (11) predicts that, at matched success probability, the useful shift is $c \leq \log(\alpha_{\max}/\alpha_0)$ and hence that the Cauchy seed ($\alpha_0 = 1$) is asymptotically the best and the An–Childs–Lin seed ($\alpha_0 = 1.31$ at $\beta = 0.7$) the worst of the three. The measured crossover matches: the β -seeded kernel (9) is cheapest down to $\epsilon_{\text{LCHS}} \approx 10^{-4}$, where the extra $e^{-b|k|^{0.7}}$ factor is still worth more than the one-norm it costs, and the Cauchy-seeded kernel takes over below that.

(iii) *Quadrature.* The FCC rule samples $\tilde{w}(\theta) = k_{\max} \cos \theta w(k_{\max} \sin \theta)$ on N_k uniform nodes of $[-\pi/2, \pi/2]$. The π -periodic extension of $\tilde{w}$ has a derivative jump of size $2k_{\max}|w(\pm k_{\max})|$ at the endpoints, so the quadrature floor at fixed N_k is set by the weight *at the truncation boundary*. For the mollified kernels that value carries the full Gaussian suppression together with all its derivatives, so the non-smooth part of the integrand is uniformly $\mathcal{O}(\epsilon)$; for the An–Childs–Lin kernel it decays only as $e^{-bk_{\max}^{0.7}}$. Figure 4 shows the effect on the sampled integrand and Fig. 2 its consequence. At $n_k = 6$ the An–Childs–Lin baseline bottoms out at $\epsilon_{\text{LCHS}} = 5.13 \times 10^{-4}$, reached at $Q_A = 600$, and

Table 5: Measured Fejér–Clenshaw–Curtis floors: the best end-to-end accuracy each family attains at each Fourier-register size anywhere on the declared $k_{\max}$ grid, and the query budget at which it occurs. Increasing $k_{\max}$ beyond that point makes the error worse, because the quadrature can no longer resolve the weight function. This is what makes the tighter rows of Table 4 unreachable at small n_k .

n_k	ACL 0.7		ACL 0.8		m-Cauchy $\hat{f}_2$		m-0.7 (ours)	
	$\epsilon_{\text{LCHS}}^{\text{floor}}$	Q_A	$\epsilon_{\text{LCHS}}^{\text{floor}}$	Q_A	$\epsilon_{\text{LCHS}}^{\text{floor}}$	Q_A	$\epsilon_{\text{LCHS}}^{\text{floor}}$	Q_A
5	1.2×10^{-3}	375	1.2×10^{-3}	375	5.3×10^{-3}	225	2.0×10^{-3}	300
6	5.1×10^{-4}	600	4.1×10^{-4}	600	3.4×10^{-4}	225	2.4×10^{-4}	525
7	4.9×10^{-5}	825	4.1×10^{-5}	750	1.7×10^{-4}	300	8.3×10^{-6}	450
8	2.1×10^{-5}	900	8.5×10^{-6}	975	6.7×10^{-6}	600	1.9×10^{-6}	750

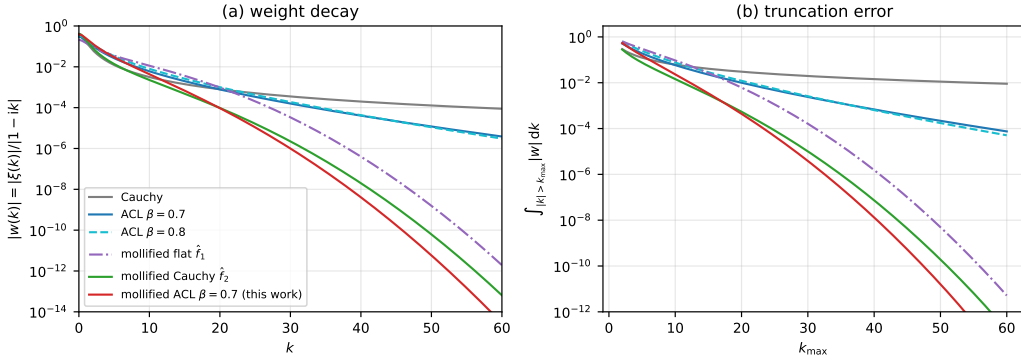


Figure 3: (a) Weight modulus for the kernels compared here, at the parameters selected for $\epsilon_{\text{LCHS}} \sim 10^{-4}$. (b) The corresponding truncation error $\int_{|k| > k_{\max}} |w| dk$. The exact kernels fall as $\exp[-bk_{\max}^\beta]$ with $\beta < 1$; the mollified kernels fall as a Gaussian at fixed γ , and their envelope over γ at a fixed one-norm budget is a pure exponential.

degrades beyond that as $k_{\max}$ grows further (it is back at 4.1×10^{-3} by $Q_A = 975$); the mollified Cauchy kernel reaches 4.88×10^{-4} at $Q_A = 150$, a factor of four fewer queries, and the mollified $\beta = 0.7$ kernel reaches 2.39×10^{-4} — twice as accurate as anything the baseline can do at that resolution — at $Q_A = 525$. At $n_k = 8$ the same comparison is 2.10×10^{-5} (baseline, $Q_A = 900$) against 1.89×10^{-6} (mollified $\beta = 0.7$, $Q_A = 750$). This is a benefit of the optimal-kernel class that, to our knowledge, has not been noted: the reference analysis predicts the same power-law quadrature region for all kernels, which is correct in form but pessimistic in amplitude for kernels whose boundary value is itself $\mathcal{O}(\epsilon)$.

5.4 Where the variant does not help

The naive log–log exponent does not separate the families. If one fits the two-parameter law $Q_A = A[\log \epsilon_{\text{LCHS}}^{-1}]^p$ — i.e. drops the additive constant that both theoretical laws carry — to the measured *envelope* over $\epsilon_{\text{LCHS}} \in [10^{-4}, 10^{-1.5}]$, the mollified kernel comes out with a *higher* exponent ($p = 1.27 \pm 4.9$ from four points) than the baseline ($p = 1.27 \pm 0.42$ from nine), even though its query count is lower at every precision in the window. The reason is visible in Fig. 1(a): the mollified curve starts from a much better intercept, so its relative growth across a fixed window is larger, and with a two-parameter fit the intercept is forced into the exponent. The fits of Table 3 use the form that both theories actually predict, $\log \epsilon_{\text{LCHS}}^{-1} = a + b k_{\max}^q$, and separate the two cleanly. We report both because the discrepancy is instructive: a naive exponent fit over a pre-asymptotic window can invert the true ordering.

Pre-asymptotic behaviour of the envelope. Relatedly, the mollified family is not in its asymptotic regime at emulator-accessible precisions. At a *fixed* (γ, c) the truncation error falls like $e^{-k_{\max}^2/4\gamma^2}$, faster than the optimal asymptotic law, until it saturates at the mollification floor $\frac{1}{2}\text{erfc}(\gamma c - \frac{1}{2\gamma})$;

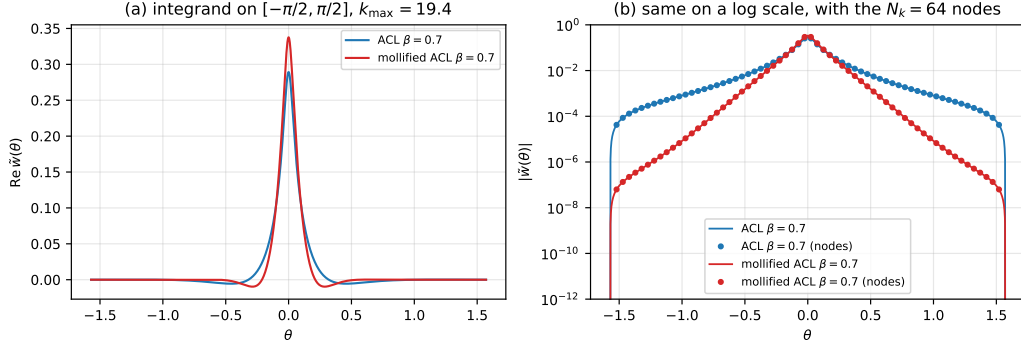


Figure 4: Why the quadrature floor differs. (a) The FCC integrand $\tilde{w}(\theta)$ at $k_{\max} = 19.4$; (b) the same integrands on a logarithmic scale, with the $N_k = 64$ nodes of the $n_k = 6$ register marked. At the truncation boundary $\theta = \pm\pi/2$ the mollified integrand is six orders of magnitude smaller than the An–Childs–Lin one, so the derivative jump that limits the FCC rule is correspondingly smaller.

the optimal $k_{\max} \propto \log \epsilon^{-1}$ law is the envelope traced by increasing γ as the target tightens. If (γ, c) are *jointly* re-optimised at every precision the optimiser also trades c downwards, and the resulting envelope is steeper than linear in $\log \epsilon^{-1}$. The fixed-kernel measurement of Table 3 avoids this ambiguity by never re-tuning the kernel between points.

Where the variant does not win. At $n_k = 5$ — only 32 Fejér–Clenshaw–Curtis nodes — the mollified kernels are the *worse* choice: their floors are 2.0×10^{-3} (mollified $\beta = 0.7$) and 5.3×10^{-3} (mollified Cauchy) against 1.2×10^{-3} for both exact kernels, a factor of 1.7 the wrong way (Table 5). The mechanism is the one of Sec. 5.3 running in reverse: with so few nodes the grid cannot resolve *any* weight function well, the quadrature error dominates before the truncation error does, and the mollified kernels’ advantage — which is entirely in the truncation error — has nothing to act on, while their narrower weight is sampled by fewer effective nodes. The crossover is sharp: by $n_k = 6$ the mollified floors are already $1.7\times$ deeper, and by $n_k = 7$ and 8 they are $4.9\times$ and $4.5\times$ deeper.

At the loosest accuracies the comparison is limited by the quantisation of $k_{\max}$ rather than by the kernels: every mollified family already meets $\epsilon_{\text{LCHS}} \approx 3 \times 10^{-3}$ at the smallest admissible bandwidth $k_{\max} = 4.55$ ($Q_A = 75$), where the An–Childs–Lin $\beta = 0.8$ kernel also does, so the measured gain there is 1.0 and is a lower bound. Among the mollified seeds, the flat seed $\hat{f}_1$ is the weakest at tight precision, exactly as Sec. 2.3 predicts: its ℓ^1 norm grows like $\log \gamma$, so part of the budget the shift needs is already spent; the doubly-rational seed $\hat{f}_{3,1}$ is worse still. And at $n_k = 8$ below $\epsilon_{\text{LCHS}} \approx 10^{-5}$ every family flattens against the quadrature floor of a 256-node grid, so further bandwidth buys nothing for any kernel — the mollified kernels simply arrive there with a factor of two fewer queries.

Kernel parameters that were tried and rejected. A super-Gaussian mollifier $\exp[-((1 + k^2)/4\gamma^2)^p]$, $p = 2$, preserves $M(-i) = 1$ and decays like e^{-k^4} , but its time-domain transform oscillates and is not positive, so the clean erfc bound of Proposition 1 is lost; we did not pursue it. Seeds with $\beta \in \{0.5, 0.6, 0.9\}$ were run and are slower than $\beta \in [0.7, 0.8]$ both with and without mollification, reproducing the near-optimal window reported for the exact family. Shifts $c > \log(\alpha_{\max}/\alpha_0)$ were rejected by the one-norm budget rather than by accuracy: they do reduce $k_{\max}$ further, but only by lowering the success probability below the baseline’s.

5.5 Classical cost

The classical work attached to a kernel substitution is small and, at these resolutions, comparable between families. Selecting $(\gamma, c, k_{\max})$ by (13) on the declared grid costs a few seconds per configuration ($\sim 10^5$ evaluations of a length- N_k weight vector); the entire declared selection sweep over four register sizes, six precision targets and eleven families took 584 s on one core. The QSVT weight construction is cheaper still: the symmetric-QSP L-BFGS solve converges in 0.1–1.3 s per parity (measured, stored in the angle files), and the Chebyshev expansion that feeds it is a few

seconds. What differs between kernels is the polynomial degree: 238 for the variant’s even part against 422 for the baseline’s, because the mollified weight modulus decays *inside* the interval and is therefore easier to approximate, while its odd part is harder (587 against 491) because the shift c adds a phase linear in k . The total number of angles is 827 for the variant against 915 for the baseline. The emulation dominates everything, but not by as much as one might fear: the 509 circuits reported here took 2.9 GPU-hours in total on H200 GPUs (measured from the sweep logs, summed in `logs/analysis/walltime.json`). The largest circuit in the study — $n_k = 8$, $Q_A = 975$, 2.1×10^5 STMC gates on 26 qubits — takes 38 s.

6 Related work

LCHS originates with An, Liu and Lin [1], whose Cauchy kernel gives $\epsilon = \mathcal{O}(k_{\max}^{-1})$, and is closely related to Schrödingerisation [8, 9]. An, Childs and Lin [2] proved the no-go result for exact kernels and introduced the near-optimal family (3), obtaining a differential-equation solver near-optimal in all parameters. The idea of replacing the exact identity by an analytic approximation goes back, in the QSP context, to de Lima Silva, Borges and Aolita [7], who approximated e^{-x} by a Fourier series obtained from an analytic extension. Low and Somma [5] and Jin, Liu, Ma, Peng and Yu [6] developed this into optimal approximate-LCHS kernels with $k_{\max} = \mathcal{O}(\log \epsilon^{-1})$; Low and Somma also give an explicit multiplexed block-encoding for the selector. Our Proposition 1 is a restatement and generalisation of their construction: their $\hat{f}_{j,y}$ family is exactly Gaussian mollification of the rational exact seeds $(\frac{y+1}{y+ik})^{j-1}/2\pi$, and (7)–(8) are their $\hat{f}_1$ and $\hat{f}_2$. We claim novelty for (a) the observation that the operation is seed-agnostic and hence applies to the An–Childs–Lin exponential seeds, giving (9); (b) the one-norm/bandwidth relation (11) and its use as the selection principle; and (c) the end-to-end implementation and measurement.

On the circuit side, Novikau and Joseph [3, 4] contributed the sinusoidal coordinate transformation, the resulting equivalence with FCC quadrature, and the single-QSP selector that we hold fixed here; their Theorem 4 states that the same construction would carry an optimal kernel to $\mathcal{O}(\log \epsilon^{-1})$ but does not test it. Related explicit constructions include the tensor-network weight preparation of Sato *et al.* [18] and the time-marching solvers of Fang, Lin and Tong [19] and Over *et al.* [20]. Qubitization and QSP/QSVT follow [11, 12, 13]; the symmetric-QSP angle solver we use for the weight function is that of Dong *et al.* [15].

7 Limitations

Regime. Everything reported here is at $n_x = 6$, $\|A_L t\| = 1.98$ and $n_k \leq 8$, i.e. a single spatial resolution and a short time. The reference implementation’s own runs use $\|A_L t\| = 127$ with $n_k = 9$ –12; we have not verified that the measured ratios persist there, although the kernel-level quantity that drives them, the truncation error $\int_{|k| > k_{\max}} |w|$, is independent of t , and the required $k_{\max}$ was found to change by less than 5% between $\|A_L t\| = 1.98$ and $\|A_L t\| = 7.94$ in the classical scans.

Compilation overhead. The selector is compiled from one shipped QSP-Hamiltonian phase sequence with $\tau_b = 39.25$ repeated n_t times. Since that sequence uses $M_b = 75$ qubitization queries for $\tau_b = 39.25$, all absolute query counts carry a common factor $M_b/\tau_b = 1.91$ relative to an optimally compiled selector. The factor is identical for every configuration and every kernel, so ratios and fitted exponents are unaffected, but the absolute Q_A values in Table 2 are about twice what a purpose-built phase sequence would need. Generating phase sequences for arbitrary τ would remove this and would also remove the quantisation of $k_{\max}$; we did not implement Haah’s completion algorithm here.

Weight construction. The headline circuits build the weights by direct decomposition, as the reference implementation does, because that needs one ancilla instead of three. The QSVT construction is computed and verified (Table 1) but is not the path exercised end to end. Its degree is modest because the mollified weight decays inside the interval; but its *odd* part is harder than the baseline’s, because the shift c adds a phase linear in k . Absorbing that phase into a spectral shift $A_L \rightarrow A_L + (c/t)I$ makes the weights real and even and halves the weight circuit, at the price of $\sim 13\%$ more selector queries at $t = 0.0125$; on the query axis chosen here that is the wrong trade, but on a gate-count or ancilla axis it may not be.

One test problem. Only the uniform advection–diffusion equation is used. Its commuting A_L, A_H are what make the operator-norm criterion (13) exact; for a non-commuting generator that criterion becomes an upper bound and the kernel selection would have to be redone with the commutator term included. The Koopman–von Neumann embedding, which is the more interesting target for LCHS and where numerical dissipation makes $\|A_L\|$ and hence $k_{\max}\|A_L\|t$ the dominant cost, is left for future work.

Statistical strength of the exponent fit. The precision exponents are fitted from 8–13 points per family on a quantised $k_{\max}$ grid, so their error bars are of order 0.1–0.3. The separation between the two families is clear in the raw data (Fig. 1) but the fitted exponents themselves should be read as consistent with, rather than a precise measurement of, the theoretical values $q = \beta$ and $q = 1$.

8 Conclusions

The step from near-optimal to optimal precision scaling in LCHS is, mathematically, one multiplication: an exact kernel times the entire mollifier $\exp[c(1 - ik) - (1 + k^2)/4\gamma^2]$, which preserves the LCHS normalisation identically, smooths the propagator’s step by a shifted Gaussian in the time variable, and carries a closed-form erfc error bound. Published optimal kernels are the rational-seed members of this family; the exponential seeds of An, Childs and Lin give a new member that is the cheapest kernel we measured down to $\epsilon_{\text{LCHS}} \approx 10^{-4}$. Inside an unmodified minimal FCC/single-QSP LCHS circuit, the substitution reduces the number of block-encoding queries by a median $2.0\times$ and up to $4\times$ at matched end-to-end accuracy and equal or better success probability, changes the measured precision law from $e^{-bk_{\max}^{0.66 \pm 0.03}}$ to $e^{-bk_{\max}^{1.14 \pm 0.04}}$, and — an effect we did not anticipate — lowers the Fejér–Clenshaw–Curtis accuracy floor at a fixed Fourier register by up to $4.9\times$, which is what makes small k -registers usable at all. That last benefit has a sharp lower edge: at $n_k = 5$, with only 32 quadrature nodes, the ordering reverses and the exact kernels floor $1.7\times$ lower, because there the quadrature error binds before the truncation error the mollifier improves. The binding resource is otherwise the ℓ^1 norm of the weights: it is what the shift spends and what the success probability pays, and it, rather than the asymptotic exponent, decides which optimal-class kernel one should use at a given precision.

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A Reproducibility

All code, inputs, emulator outputs and logs are in `proposal/`. The pre-registered plan is `proposal/code/kernel_plan.{md,json,py}`; the kernel derivation and its numerical verification are in `proposal/code/kernel_derivation/`; the emulator inputs, raw `.hdf5` outputs and per-run metrics for every circuit reported here are in `proposal/logs/runs/`. `proposal/codebase_notes.md` records what was inspected, reused and extended in the provided framework, and `proposal/attempts_log.md` records the approaches that were tried and rejected.